

INTEGER SOLUTIONS OF $y^2 + x^2y + xz^2 + 2 = 0$

ABSTRACT. We determine all integer triples (x, y, z) satisfying $y^2 + x^2y + xz^2 + 2 = 0$. The principal result is an exact divisor–square parametrization in which the parameter $|x|$ is recovered uniquely from the solution. We then give a complete fixed- $|x|$ description: the positive- x branch is finite, the negative branch with square $|x|$ is governed by a finite divisor calculation, and every solvable negative branch with nonsquare $|x|$ is a finite disjoint union of explicit Pell recurrences. We also prove the precise congruence obstruction on $|x|$, a global unimodular parametrization, and an explicit infinite family with $x = -2$.

1. STATEMENT OF THE CLASSIFICATION

We study the integral points on

$$(1.1) \quad y^2 + x^2y + xz^2 + 2 = 0, \quad x, y, z \in \mathbb{Z}.$$

Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$. For $m \in \mathbb{N}$ and $y \in \mathbb{Z}$ such that $m \mid y^2 + 2$, define

$$(1.2) \quad F_m(y) := my + \frac{y^2 + 2}{m}.$$

The following theorem is the basic complete parametrization.

Theorem 1.1 (Exact divisor–square parametrization). *The integer solutions of (1.1) are exactly the triples*

$$(1.3) \quad (x, y, z) = (-\varepsilon m, y, \nu t),$$

where

$$(1.4) \quad m, t \in \mathbb{N}, \quad y \in \mathbb{Z}, \quad m \mid y^2 + 2, \quad F_m(y) = \varepsilon t^2, \quad \varepsilon, \nu \in \{\pm 1\}.$$

For every solution, the parameters $m = |x|$, y , $t = |z|$, and $\varepsilon = -\operatorname{sgn}(x)$ are uniquely determined; only $\nu = \operatorname{sgn}(z)$ records the choice of sign of z . In particular, $F_m(y)$ never vanishes for an admissible pair (m, y) occurring in a solution.

Thus an equivalent concise description is

$$(1.5) \quad (x, y, z) = \left(-\operatorname{sgn}(F_m(y))m, y, \pm \sqrt{|F_m(y)|} \right),$$

where $m \mid y^2 + 2$ and $|F_m(y)|$ is a perfect square. The nonvanishing assertion in Theorem 1.1 makes (1.5) unambiguous.

The later sections refine Theorem 1.1 in three ways. First, we characterize exactly the positive integers m for which the necessary congruence $y^2 \equiv -2 \pmod{m}$ is soluble. Second, for each fixed m we convert the remaining square condition into a positive-definite equation, a finite factorization, or a finite union of Pell orbits according to the sign of x and whether m is a square. Third, we give a global parametrization by matrices in $\operatorname{SL}_2(\mathbb{Z})$.

2. PROOF OF THE EXACT PARAMETRIZATION

We begin by excluding the two degeneracies that could otherwise cause ambiguity.

Lemma 2.1. *Every solution of (1.1) satisfies $x \neq 0$ and $z \neq 0$.*

Proof. If $x = 0$, then (1.1) becomes $y^2 + 2 = 0$, which has no integer solution.

Suppose next that $z = 0$. Since $x \neq 0$, equation (1.1) becomes

$$(2.1) \quad y(y + x^2) = -2.$$

Because $x^2 > 0$, the second factor is strictly larger than the first. The only ordered pairs of integral factors of -2 with this property are $(-2, 1)$ and $(-1, 2)$. In either case their difference is 3, so (2.1) would force $x^2 = 3$, impossible. Therefore $z \neq 0$. $\square$

Proof of Theorem 1.1. Let (x, y, z) satisfy (1.1). By Lemma 2.1, we may write

$$(2.2) \quad x = \delta m, \quad m = |x| \in \mathbb{N}, \quad \delta \in \{\pm 1\}.$$

Then $x^2 = m^2$, and (1.1) is

$$(2.3) \quad y^2 + m^2 y + \delta m z^2 + 2 = 0.$$

Reduction modulo m gives

$$(2.4) \quad y^2 + 2 \equiv 0 \pmod{m}.$$

Consequently

$$(2.5) \quad q := \frac{y^2 + 2}{m}$$

is a positive integer. Dividing (2.3) by m yields

$$(2.6) \quad q + m y + \delta z^2 = 0.$$

By (1.2), this says

$$(2.7) \quad F_m(y) = -\delta z^2.$$

Set $t = |z|$, $\varepsilon = -\delta$, and $\nu = \operatorname{sgn}(z)$. Lemma 2.1 gives $t \in \mathbb{N}$, and (2.7) becomes $F_m(y) = \varepsilon t^2$. Moreover $x = \delta m = -\varepsilon m$ and $z = \nu t$, proving that every solution occurs in (1.3)–(1.4).

Conversely, suppose the data in (1.4) are given and set (x, y, z) by (1.3). Since $x^2 = m^2$ and $z^2 = t^2$, we have

$$\begin{aligned} y^2 + x^2 y + x z^2 + 2 &= y^2 + m^2 y - \varepsilon m t^2 + 2 \\ &= m \left(\frac{y^2 + 2}{m} + m y - \varepsilon t^2 \right) \\ &= m (F_m(y) - \varepsilon t^2) = 0. \end{aligned}$$

Thus every parameter choice in the theorem gives a solution.

For a given solution, $m = |x|$, the displayed value of y , and $t = |z|$ are forced. Equation (2.7) then forces $\varepsilon = -\operatorname{sgn}(x)$, while ν is exactly the sign of z . Finally, $F_m(y) = \varepsilon t^2 \neq 0$ because $t \in \mathbb{N}$. $\square$

The two signs of x lead to useful specialized forms.

Corollary 2.2 (The positive branch). *The solutions with $x > 0$ are exactly*

$$(2.8) \quad (x, y, z) = \left(m, -a, \pm \sqrt{ma - \frac{a^2 + 2}{m}} \right),$$

where

$$(2.9) \quad m \in \mathbb{N}, \quad 1 \leq a \leq m^2 - 1, \quad m \mid a^2 + 2,$$

and

$$(2.10) \quad ma - \frac{a^2 + 2}{m}$$

is a positive perfect square. For a fixed m , the substitution

$$(2.11) \quad a \mapsto m^2 - a$$

preserves the value in (2.10).

Proof. Put $x = m > 0$ in (1.1). Then

$$(2.12) \quad y(y + m^2) = -m z^2 - 2 < 0.$$

The two factors on the left differ by the positive integer m^2 , so they have opposite signs exactly when $-m^2 < y < 0$. Write $a = -y$; then $1 \leq a \leq m^2 - 1$. Theorem 1.1, with $F_m(-a) < 0$, gives

$$z^2 = -F_m(-a) = ma - \frac{a^2 + 2}{m},$$

which proves the parametrization and the square condition. Conversely, these conditions are a specialization of Theorem 1.1 and therefore produce a solution.

Finally,

$$\begin{aligned} m(m^2 - a) - \frac{(m^2 - a)^2 + 2}{m} &= m^3 - ma - \left(m^3 - 2ma + \frac{a^2 + 2}{m}\right) \\ &= ma - \frac{a^2 + 2}{m}. \end{aligned}$$

Also $(m^2 - a)^2 + 2 \equiv a^2 + 2 \pmod{m}$, so divisibility is preserved. This proves the final assertion. $\square$

Example 2.3 (Some positive branches). *Corollary 2.2 gives the following complete lists for three small positive values of x :*

$$\begin{aligned} x = 2 : \quad & (y, z) = (-2, \pm 1), \\ x = 3 : \quad & (y, z) = (-2, \pm 2), (-7, \pm 2), \\ x = 9 : \quad & (y, z) = (-22, \pm 12), (-59, \pm 12). \end{aligned}$$

For instance, when $m = 9$ the only admissible values of a for which (2.10) is a square are $a = 22$ and $a = 59$; the involution $a \mapsto 81 - a$ interchanges them.

Corollary 2.4 (The negative branch). *The solutions with $x < 0$ are exactly*

$$(2.13) \quad (x, y, z) = \left(-m, y, \pm \sqrt{my + \frac{y^2 + 2}{m}}\right),$$

where $m \in \mathbb{N}$, $m \mid y^2 + 2$, and

$$(2.14) \quad my + \frac{y^2 + 2}{m}$$

is a positive perfect square.

Proof. This is Theorem 1.1 with $\varepsilon = 1$, equivalently with $x = -m$. $\square$

The quadratic nature of (1.1) in y gives an elementary symmetry that is occasionally useful.

Proposition 2.5 (Vieta involution). *If (x, y, z) satisfies (1.1), then so does*

$$(2.15) \quad (x, -x^2 - y, z).$$

This transformation is an involution. On the parameters of Corollary 2.2, it is exactly $a \mapsto m^2 - a$.

Proof. For fixed x, z , equation (1.1) is the monic quadratic

$$Y^2 + x^2 Y + (xz^2 + 2) = 0.$$

The sum of its two roots is $-x^2$. Hence, if one root is y , the other is $-x^2 - y$. Applying the transformation twice returns y , so it is an involution. If $x = m > 0$ and $y = -a$, the new value is $-m^2 + a = -(m^2 - a)$, as claimed. $\square$

3. THE EXACT CONGRUENCE OBSTRUCTION ON $|x|$

The divisibility condition in Theorem 1.1 can itself be classified completely.

For an odd prime p and an integer a not divisible by p , write $\left(\frac{a}{p}\right) = 1$ when a is a quadratic residue modulo p and $\left(\frac{a}{p}\right) = -1$ otherwise.

Lemma 3.1 (Gauss's lemma). *Let p be an odd prime, let $a \not\equiv 0 \pmod{p}$, and put $n = (p - 1)/2$. For $1 \leq j \leq n$, let r_j be the unique integer with*

$$-\frac{p}{2} < r_j < \frac{p}{2}, \quad r_j \equiv aj \pmod{p}.$$

If exactly N of the integers r_j are negative, then

$$(3.1) \quad \left(\frac{a}{p}\right) = (-1)^N.$$

Proof. The absolute values $|r_1|, \dots, |r_n|$ are distinct. Indeed, if $|r_i| = |r_j|$, then $ai \equiv \pm aj \pmod{p}$, hence $i \equiv \pm j \pmod{p}$. In the ranges $1 \leq i, j \leq n$, the plus sign forces $i = j$, while the minus sign would give $i + j = p$, impossible because $i + j \leq p - 1$. Therefore the $|r_j|$ are a permutation of $1, 2, \dots, n$. Taking products modulo p gives

$$a^n n! \equiv \prod_{j=1}^n r_j = (-1)^N \prod_{j=1}^n |r_j| = (-1)^N n! \pmod{p}.$$

Since $p \nmid n!$, cancellation yields $a^n \equiv (-1)^N \pmod{p}$.

It remains only to identify a^n with the Legendre symbol. Fermat's little theorem follows because multiplication by a permutes the nonzero residue classes modulo p :

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p},$$

so $a^{p-1} \equiv 1 \pmod{p}$. The n nonzero squares modulo p are distinct and are roots of $X^n - 1$; since a nonzero polynomial of degree n over a field has at most n roots, these are all the roots. Thus $a^n \equiv 1 \pmod{p}$ exactly when a is a square. If a is not a square, then $a^n \not\equiv 1 \pmod{p}$ while $(a^n)^2 = a^{p-1} \equiv 1 \pmod{p}$, so $a^n \equiv -1 \pmod{p}$. Consequently $a^n \equiv \left(\frac{a}{p}\right) \pmod{p}$. Both sides of (3.1) belong to $\{\pm 1\}$, and the lemma follows. The same criterion also proves multiplicativity: for $a, b \not\equiv 0 \pmod{p}$,

$$\left(\frac{ab}{p}\right) \equiv (ab)^n \equiv a^n b^n \equiv \left(\frac{a}{p}\right) \left(\frac{b}{p}\right) \pmod{p},$$

and both sides are again in $\{\pm 1\}$. □

Lemma 3.2 (The residue of -2 modulo an odd prime). *Let p be an odd prime. Then*

$$(3.2) \quad u^2 \equiv -2 \pmod{p}$$

has a solution if and only if

$$(3.3) \quad p \equiv 1 \quad \text{or} \quad 3 \pmod{8}.$$

When it is soluble, it has exactly two roots modulo p .

Proof. By Lemma 3.1, applied first to $a = -1$ and then to $a = 2$, we obtain

$$(3.4) \quad \left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}, \quad \left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}.$$

For completeness, the second count is as follows. The least positive residues of $2, 4, \dots, p-1$ exceed $p/2$ precisely for the integers j with $p/4 < j \leq (p-1)/2$. Their number is

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor,$$

whose parity, checked in the four odd classes modulo 8, is the parity of $(p^2-1)/8$. The first formula is immediate because all $(p-1)/2$ residues of $-j$ are negative in least absolute form.

By multiplicativity of the Legendre symbol,

$$(3.5) \quad \left(\frac{-2}{p}\right) = (-1)^{(p-1)/2 + (p^2-1)/8}.$$

Checking the four odd residue classes modulo 8, the exponent in (3.5) is even exactly for $p \equiv 1, 3 \pmod{8}$. Thus -2 is a quadratic residue exactly in those two classes. If a root u exists, then $-u$ is a distinct root because $p \neq 2$, and a nonzero quadratic polynomial over the field $\mathbb{Z}/p\mathbb{Z}$ has at most two roots. Hence there are exactly two. □

Lemma 3.3 (Lifting roots through odd prime powers). *Let p be an odd prime. Every solution of*

$$(3.6) \quad u^2 \equiv -2 \pmod{p^k}, \quad k \geq 1,$$

lifts uniquely modulo p^{k+1} to a solution of the same congruence. Consequently, if -2 is a quadratic residue modulo p , then $u^2 \equiv -2 \pmod{p^e}$ has exactly two roots for every $e \geq 1$.

Proof. Write $u^2 + 2 = p^k c$. A lift has the form $u + tp^k$, with t considered modulo p . Since $2k \geq k + 1$,

$$(u + tp^k)^2 + 2 \equiv p^k(c + 2ut) \pmod{p^{k+1}}.$$

The residue of u modulo p is nonzero, because $u^2 \equiv -2 \not\equiv 0 \pmod{p}$. Hence $2u$ is invertible modulo p , and there is a unique $t \pmod{p}$ satisfying $c + 2ut \equiv 0 \pmod{p}$. This proves unique lifting. Starting from the two roots modulo p supplied by Lemma 3.2 and iterating gives exactly two roots modulo every p^e . $\square$

Theorem 3.4 (Soluble moduli). *For $m \in \mathbb{N}$, the congruence*

$$(3.7) \quad y^2 \equiv -2 \pmod{m}$$

is soluble if and only if

$$(3.8) \quad m = 2^\epsilon \prod_{i=1}^r p_i^{e_i}, \quad \epsilon \in \{0, 1\}, \quad p_i \equiv 1 \text{ or } 3 \pmod{8},$$

where the p_i are distinct odd primes and $e_i \in \mathbb{N}$. For such an m , the congruence has exactly 2^r residue classes of solutions modulo m .

Proof. Suppose first that (3.7) is soluble. If $4 \mid m$, then it would imply $y^2 \equiv 2 \pmod{4}$, impossible because a square modulo 4 is 0 or 1. Thus the exponent of 2 in m is at most one. If an odd prime $p \mid m$, reduction modulo p shows that -2 is a quadratic residue modulo p . By Lemma 3.2, $p \equiv 1$ or $3 \pmod{8}$. This proves the necessity of (3.8).

Conversely, assume (3.8). For each odd prime power $p_i^{e_i}$, Lemmas 3.2 and 3.3 provide exactly two roots. Modulo 2, the unique root is 0. The moduli $2^\epsilon, p_1^{e_1}, \dots, p_r^{e_r}$ are pairwise coprime, so the Chinese remainder theorem combines an arbitrary choice of one root for each odd prime power into a unique root modulo m . There are therefore exactly 2^r roots.

For completeness, the Chinese remainder theorem in this setting follows directly from Bezout's identity: for pairwise coprime moduli n_j , put $N = \prod_j n_j$, $N_j = N/n_j$, and choose $c_j N_j \equiv 1 \pmod{n_j}$; then $\sum_j a_j c_j N_j$ has the prescribed residue a_j modulo each n_j , and uniqueness holds modulo N . $\square$

Corollary 3.5 (Restrictions on x, y, z). *If (x, y, z) satisfies (1.1), then*

$$(3.9) \quad |x| = 2^\epsilon \prod_{i=1}^r p_i^{e_i}$$

with the restrictions in Theorem 3.4. Moreover,

$$(3.10) \quad \begin{aligned} |x| \text{ odd} &\implies z \text{ even}, \\ |x| \text{ even} &\implies y \text{ even and } z \text{ odd}. \end{aligned}$$

Proof. The factorization restriction follows from (2.4) and Theorem 3.4.

If $|x|$ is odd, then reduction of (1.1) modulo 2 gives

$$y^2 + y + z^2 \equiv 0 \pmod{2}.$$

Since $y^2 + y$ is always even, z^2 is even, and hence z is even.

If $|x|$ is even, Theorem 3.4 shows that its exact 2-adic valuation is one, so $x \equiv 2 \pmod{4}$. Congruence (2.4) modulo 2 forces y to be even. Reducing (1.1) modulo 4, the terms y^2 and $x^2 y$ vanish, and we obtain

$$2z^2 + 2 \equiv 0 \pmod{4}.$$

Thus $z^2 \equiv 1 \pmod{2}$, so z is odd. $\square$

4. REDUCTION TO QUADRATIC NORM EQUATIONS

The remaining square condition in Theorem 1.1 has a useful symmetric form.

Proposition 4.1 (Quadratic norm form). *Let $m \in \mathbb{N}$, $\delta \in \{\pm 1\}$, and put $x = \delta m$. There is a bijection between solutions (x, y, z) of (1.1) with this fixed value of x and integer pairs (U, V) satisfying*

$$(4.1) \quad U^2 + \delta m V^2 = m^4 - 8,$$

together with the parity conditions

$$(4.2) \quad U \equiv m^2 \pmod{2}, \quad V \equiv 0 \pmod{2}.$$

The bijection is

$$(4.3) \quad U = 2y + m^2, \quad V = 2z,$$

and

$$(4.4) \quad y = \frac{U - m^2}{2}, \quad z = \frac{V}{2}.$$

Proof. With $x = \delta m$, equation (1.1) is

$$y^2 + m^2 y + \delta m z^2 + 2 = 0.$$

Multiplying by 4 and completing the square in y gives

$$(2y + m^2)^2 + \delta m (2z)^2 = m^4 - 8,$$

which is (4.1). The definitions in (4.3) imply (4.2).

Conversely, the parity conditions make the quantities in (4.4) integers. Substituting them into (4.1) and reversing the completion of the square gives (1.1). The two constructions are inverse to one another. $\square$

4.1. Positive x . For $x = m > 0$, the norm equation is positive definite.

Corollary 4.2 (Finiteness for fixed positive x). *For every fixed positive integer m , there are only finitely many solutions with $x = m$. More precisely, there are no such solutions for $m = 1$, and for $m \geq 2$ every corresponding pair (U, V) satisfies*

$$(4.5) \quad |U| \leq \sqrt{m^4 - 8}, \quad |V| \leq \sqrt{\frac{m^4 - 8}{m}}.$$

Checking the finitely many pairs in this box subject to (4.2) gives all solutions.

Proof. When $m = 1$, equation (4.1) is $U^2 + V^2 = -7$, impossible. For $m \geq 2$, it is

$$U^2 + m V^2 = m^4 - 8 > 0.$$

Both summands are nonnegative, giving the two bounds in (4.5). A bounded subset of $\mathbb{Z}^2$ is finite, and Proposition 4.1 completes the proof. $\square$

4.2. Negative x with square $|x|$. When $x = -m$ and m is a square, the norm equation factors over $\mathbb{Z}$.

Theorem 4.3 (Finite divisor parametrization for $x = -d^2$). *Fix $d \in \mathbb{N}$. The solutions of (1.1) with $x = -d^2$ are in bijection with ordered pairs $(A, B) \in \mathbb{Z}^2$ satisfying*

$$(4.6) \quad AB = d^8 - 8,$$

$$(4.7) \quad A + B \equiv 2d^4 \pmod{4}, \quad B - A \equiv 0 \pmod{4d}.$$

The corresponding solution is

$$(4.8) \quad (x, y, z) = \left(-d^2, \frac{A + B - 2d^4}{4}, \frac{B - A}{4d} \right).$$

In particular, for each square value of $|x|$, there are only finitely many solutions.

Proof. Set $m = d^2$ and $\delta = -1$ in Proposition 4.1. The norm equation becomes

$$(4.9) \quad U^2 - d^2 V^2 = d^8 - 8.$$

Define

$$(4.10) \quad A = U - dV, \quad B = U + dV.$$

Then (4.9) is exactly (4.6). Since $U = 2y + d^4$ and $V = 2z$, we also have

$$A + B = 4y + 2d^4, \quad B - A = 4dz,$$

which proves the congruences in (4.7) and the recovery formulas in (4.8).

Conversely, suppose (A, B) satisfies (4.6)–(4.7). Put

$$U = \frac{A + B}{2}, \quad V = \frac{B - A}{2d}.$$

The first congruence in (4.7) implies that U is an integer with $U \equiv d^4 = m^2 \pmod{2}$; the second implies that V is an even integer. Equation (4.6) gives $U^2 - d^2 V^2 = d^8 - 8$. Proposition 4.1 therefore produces the solution (4.8). The two constructions are inverse. Finally, the nonzero integer $d^8 - 8$ has finitely many ordered factor pairs, proving finiteness. $\square$

Example 4.4. For $d = 1$, Theorem 4.3 gives exactly

$$(4.11) \quad (x, y, z) = (-1, 1, \pm 2), \quad (-1, -2, \pm 2).$$

Indeed, in this case (A, B) runs through the admissible ordered factor pairs of -7 .

4.3. Negative x with nonsquare $|x|$. We first prove the elementary Pell fact needed to make the description self-contained.

Lemma 4.5 (Existence of a positive Pell unit). *If $m \in \mathbb{N}$ is not a square, then there exist $a, b \in \mathbb{N}$ such that*

$$(4.12) \quad a^2 - mb^2 = 1.$$

Proof. For every integer $H \geq 2$, apply the pigeonhole principle to the $H + 1$ fractional parts

$$0, \{\sqrt{m}\}, \{2\sqrt{m}\}, \dots, \{H\sqrt{m}\}$$

placed in H half-open intervals of length $1/H$. Two lie in the same interval. Subtracting their indices gives integers p, q with $1 \leq q \leq H$ and

$$(4.13) \quad |q\sqrt{m} - p| < \frac{1}{H}.$$

Because m is not a square,

$$(4.14) \quad c := p^2 - mq^2$$

is a nonzero integer. Also

$$(4.15) \quad \begin{aligned} |c| &= |p - q\sqrt{m}| |p + q\sqrt{m}| \\ &< \frac{1}{H} (2H\sqrt{m} + 1) < 2\sqrt{m} + 1. \end{aligned}$$

The pairs (p, q) obtained as H varies cannot belong to a finite set: if they did, the positive numbers $|q\sqrt{m} - p|$ arising from that finite set would have a positive minimum, contradicting (4.13) for sufficiently large H . Thus infinitely many distinct pairs occur. Since (4.15) leaves only finitely many possible nonzero values of c , one fixed value of c occurs for infinitely many pairs. Passing to a further infinite subsequence, we may assume that p and q have fixed residue classes modulo $|c|$.

For all sufficiently large members of this subsequence, $p > 0$. Choose two distinct pairs (p_1, q_1) , (p_2, q_2) from it such that

$$p_2 + q_2\sqrt{m} > p_1 + q_1\sqrt{m} > 0.$$

Because the two pairs are congruent componentwise modulo $|c|$, the integers

$$(4.16) \quad A = \frac{p_2 p_1 - m q_2 q_1}{c}, \quad B = \frac{q_2 p_1 - p_2 q_1}{c}$$

are well defined. Indeed, modulo c the first numerator is $p_1^2 - mq_1^2 = c$, and the second is zero. In the real quadratic field,

$$(4.17) \quad A + B\sqrt{m} = \frac{p_2 + q_2\sqrt{m}}{p_1 + q_1\sqrt{m}} > 1.$$

The norm of this quotient is $c/c = 1$. Its conjugate is therefore the positive number $(A + B\sqrt{m})^{-1} < 1$. Hence

$$A = \frac{(A + B\sqrt{m}) + (A - B\sqrt{m})}{2} > 0,$$

and

$$B\sqrt{m} = \frac{(A + B\sqrt{m}) - (A - B\sqrt{m})}{2} > 0.$$

Thus $A, B \in \mathbb{N}$, and taking norms in (4.17) gives $A^2 - mB^2 = 1$. Renaming (A, B) as (a, b) proves the lemma. $\square$

Theorem 4.6 (Complete Pell-orbit description). *Fix a nonsquare integer $m \geq 2$, and set*

$$(4.18) \quad N_m = m^4 - 8, \quad R_m = \sqrt{N_m}.$$

Choose $a, b \in \mathbb{N}$ with $a^2 - mb^2 = 1$, and write

$$(4.19) \quad \eta = (a + b\sqrt{m})^2 = P + Q\sqrt{m}.$$

Then

$$(4.20) \quad P^2 - mQ^2 = 1, \quad P \text{ is odd}, \quad Q \text{ is even}, \quad \eta > 1.$$

Define the seed set

$$(4.21) \quad \mathcal{B}_m := \{(U, V) \in \mathbb{Z}^2 : U^2 - mV^2 = N_m, \quad U \equiv m^2 \pmod{2}, \quad V \equiv 0 \pmod{2}, \\ R_m \leq |U + V\sqrt{m}| < \eta R_m\}.$$

Then $\mathcal{B}_m$ is finite, and every solution of (1.1) with $x = -m$ is obtained uniquely by choosing a seed $(U_0, V_0) \in \mathcal{B}_m$, an integer $n \in \mathbb{Z}$, and setting

$$(4.22) \quad U_n + V_n\sqrt{m} = (U_0 + V_0\sqrt{m})\eta^n,$$

followed by

$$(4.23) \quad (x, y, z) = \left(-m, \frac{U_n - m^2}{2}, \frac{V_n}{2}\right).$$

Equivalently, consecutive elements of each orbit satisfy

$$(4.24) \quad \begin{aligned} U_{n+1} &= PU_n + mQV_n, \\ V_{n+1} &= QU_n + PV_n. \end{aligned}$$

Consequently, for every fixed nonsquare m , the equation with $x = -m$ has either no integer solutions or infinitely many integer solutions.

Proof. Lemma 4.5 supplies a, b . Expanding (4.19) gives

$$P = a^2 + mb^2 = 1 + 2mb^2, \quad Q = 2ab.$$

Thus P is odd, Q is even, and $P, Q > 0$. The norm of η is $(a^2 - mb^2)^2 = 1$, proving (4.20).

We next prove that $\mathcal{B}_m$ is finite. For a seed put

$$\alpha = U + V\sqrt{m}, \quad \bar{\alpha} = U - V\sqrt{m}.$$

Its norm is $N_m > 0$. Since $|\alpha| \geq R_m$,

$$|\bar{\alpha}| = \frac{N_m}{|\alpha|} \leq R_m.$$

Together with $|\alpha| < \eta R_m$, this gives

$$(4.25) \quad |U| < \frac{\eta + 1}{2} R_m, \quad |V| < \frac{\eta + 1}{2\sqrt{m}} R_m.$$

Indeed, $2U = \alpha + \bar{\alpha}$ and $2V\sqrt{m} = \alpha - \bar{\alpha}$. The bounds (4.25) place all seeds in a bounded rectangle in $\mathbb{Z}^2$, so there are finitely many.

Multiplication by η preserves the norm equation because $P^2 - mQ^2 = 1$. In coordinates it is exactly the recurrence (4.24). Since P is odd and Q is even,

$$U_{n+1} \equiv U_n \pmod{2}, \quad V_{n+1} \equiv V_n \pmod{2}.$$

The inverse unit is $\eta^{-1} = P - Q\sqrt{m}$, so the same conclusions hold for negative powers. Thus every orbit in (4.22) consists entirely of parity-compatible solutions of the norm equation. Proposition 4.1 then gives (4.23).

Conversely, let (U, V) be any parity-compatible solution of $U^2 - mV^2 = N_m$, and put $\alpha = U + V\sqrt{m} \neq 0$. Since $\eta > 1$, there is a unique integer k such that

$$(4.26) \quad R_m \leq |\alpha|\eta^k < \eta R_m.$$

For example, one may take

$$k = -\lfloor \log_\eta(|\alpha|/R_m) \rfloor,$$

for then $0 \leq \log_\eta(|\alpha|/R_m) + k < 1$, which is exactly (4.26). The element $\alpha\eta^k$ still belongs to $\mathbb{Z}[\sqrt{m}]$, has norm N_m , and satisfies the required parity conditions. It therefore corresponds to a unique seed in $\mathcal{B}_m$. Reversing the power shows that (U, V) lies in the orbit of that seed. The half-open interval in (4.26) also proves uniqueness of the seed and exponent.

If $\mathcal{B}_m$ is empty, there are no solutions. If it contains a seed α_0 , the elements $\alpha_0\eta^n$, $n \in \mathbb{Z}$, are pairwise distinct because $\eta > 1$ and $\alpha_0 \neq 0$. Hence there are infinitely many solutions. $\square$

5. A GLOBAL UNIMODULAR PARAMETRIZATION

The divisor-square parametrization can be encoded by the unique positive-definite integral binary quadratic form of discriminant -8 . We include the reduction argument in full.

Lemma 5.1 (Forms of discriminant -8). *Every positive-definite integral binary quadratic form*

$$(5.1) \quad f(S, T) = AS^2 + BST + CT^2$$

with discriminant

$$(5.2) \quad B^2 - 4AC = -8$$

is properly equivalent over $\mathbb{Z}$ to

$$(5.3) \quad S^2 + 2T^2.$$

That is, some matrix in $\mathrm{SL}_2(\mathbb{Z})$ transforms one form into the other.

Proof. Because f is positive definite, it assumes a least positive value on $\mathbb{Z}^2 \setminus \{(0, 0)\}$; denote this value by A_0 , attained at a vector e_1 . The vector e_1 is primitive, since otherwise $e_1 = de'_1$ with $|d| > 1$ would give $f(e'_1) = f(e_1)/d^2 < A_0$. By Bezout's identity, e_1 extends to an oriented integral basis (e_1, e_2) of $\mathbb{Z}^2$.

In this basis the form is

$$A_0S^2 + B_0ST + C_0T^2.$$

Replace e_2 by $e_2 + ke_1$, choosing $k \in \mathbb{Z}$ nearest to $-B_0/(2A_0)$. This basis change has determinant one, leaves the first coefficient equal to A_0 , and changes the middle coefficient to

$$B_1 = B_0 + 2kA_0$$

with $|B_1| \leq A_0$. Let the new last coefficient be C_1 . Since the new second basis vector is nonzero and integral, the minimality of A_0 gives $C_1 \geq A_0$. The discriminant remains -8 , so

$$8 = 4A_0C_1 - B_1^2 \geq 4A_0^2 - A_0^2 = 3A_0^2.$$

As A_0 is a positive integer, this forces $A_0 = 1$. Equation (5.2) also implies that B_1 is even. Since $|B_1| \leq 1$, we get $B_1 = 0$, and then $4C_1 = 8$, so $C_1 = 2$. All basis changes used had determinant one, proving proper equivalence to (5.3). $\square$

Theorem 5.2 (Unimodular parametrization). *Let*

$$(5.4) \quad G = \begin{pmatrix} r & u \\ s & v \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}), \quad rv - su = 1,$$

and define

$$(5.5) \quad M = r^2 + 2s^2, \quad Y = ru + 2sv, \quad Q = u^2 + 2v^2,$$

and

$$(5.6) \quad T = MY + Q.$$

Then

$$(5.7) \quad MQ - Y^2 = 2$$

and $T \neq 0$. Whenever $|T| = t^2$ is a perfect square, necessarily $t \in \mathbb{N}$, and

$$(5.8) \quad (x, y, z) = (-\mathrm{sgn}(T)M, Y, \pm t)$$

is a solution of (1.1).

Conversely, every integer solution of (1.1) is obtained from at least one matrix $G \in \mathrm{SL}_2(\mathbb{Z})$ in this way.

Proof. Expanding the left side of (5.7) gives

$$\begin{aligned} MQ - Y^2 &= (r^2 + 2s^2)(u^2 + 2v^2) - (ru + 2sv)^2 \\ &= 2(r^2v^2 + s^2u^2 - 2rsuv) \\ &= 2(rv - su)^2 = 2. \end{aligned}$$

Suppose for contradiction that $T = MY + Q = 0$. Since $M, Q > 0$, we have $Y < 0$. Substitution of $Q = -MY$ into (5.7) gives

$$-Y(M^2 + Y) = 2.$$

Writing $a = -Y > 0$, this becomes

$$a(M^2 - a) = 2.$$

Both factors are positive integers. They must be $(1, 2)$ or $(2, 1)$, and either case gives $M^2 = 3$, impossible. Hence $T \neq 0$.

Now suppose $|T| = t^2$. Since $T \neq 0$, $t \in \mathbb{N}$. For either choice of the sign of z , set (x, y, z) as in (5.8). Then $x^2 = M^2$, $z^2 = t^2$, and $\mathrm{sgn}(T)t^2 = T$. Therefore

$$\begin{aligned} y^2 + x^2y + xz^2 + 2 &= Y^2 + M^2Y - MT + 2 \\ &= Y^2 + M^2Y - M(MY + Q) + 2 \\ &= Y^2 - MQ + 2 = 0 \end{aligned}$$

by (5.7).

For completeness, let (x, y, z) be any solution. Put

$$m = |x|, \quad q = \frac{y^2 + 2}{m}.$$

Theorem 1.1 shows that $q \in \mathbb{N}$, and

$$(5.9) \quad mq - y^2 = 2.$$

Consider the positive-definite integral form

$$(5.10) \quad f(S, T) = mS^2 + 2yST + qT^2.$$

Its discriminant is $4y^2 - 4mq = -8$. By Lemma 5.1, there is a matrix $G \in \mathrm{SL}_2(\mathbb{Z})$ such that

$$(5.11) \quad f(S, T) = (rS + uT)^2 + 2(sS + vT)^2.$$

Comparing coefficients in (5.11) gives

$$m = M, \quad y = Y, \quad q = Q.$$

The linearized equation (2.6) says

$$T = MY + Q = my + q = -\operatorname{sgn}(x)z^2.$$

Thus $|T| = z^2$ and $-\operatorname{sgn}(T)M = x$. The given solution is therefore recovered from (5.8). $\square$

The dependence on the second column of G can be written as one explicit quadratic polynomial.

Corollary 5.3 (One-variable form of the square condition). *Fix a primitive pair $(r, s) \in \mathbb{Z}^2$, choose one pair $(u_0, v_0) \in \mathbb{Z}^2$ with*

$$(5.12) \quad rv_0 - su_0 = 1,$$

and put

$$(5.13) \quad M = r^2 + 2s^2, \quad A = ru_0 + 2sv_0, \quad Q_0 = u_0^2 + 2v_0^2.$$

Every second column $(u, v)^T$ completing $(r, s)^T$ to a matrix in $\operatorname{SL}_2(\mathbb{Z})$ is uniquely of the form

$$(5.14) \quad u = u_0 + kr, \quad v = v_0 + ks, \quad k \in \mathbb{Z}.$$

For this matrix, the signed square T in Theorem 5.2 is

$$(5.15) \quad T(k) = Mk^2 + (M^2 + 2A)k + MA + Q_0,$$

and the discriminant of this quadratic polynomial is

$$(5.16) \quad \operatorname{disc} T = M^4 - 8.$$

Proof. If another pair (u, v) satisfies $rv - su = 1$, subtracting (5.12) gives

$$r(v - v_0) = s(u - u_0).$$

Since $\gcd(r, s) = 1$, there is a unique $k \in \mathbb{Z}$ with $u - u_0 = kr$ and $v - v_0 = ks$. This proves (5.14).

For these values,

$$Y = A + kM, \quad Q = Q_0 + 2kA + k^2M.$$

Hence

$$\begin{aligned} T(k) &= MY + Q = M(A + kM) + Q_0 + 2kA + k^2M \\ &= Mk^2 + (M^2 + 2A)k + MA + Q_0, \end{aligned}$$

which is (5.15). Finally, the identity $MQ_0 - A^2 = 2$, which is (5.7) for the initial matrix, gives

$$\begin{aligned} \operatorname{disc} T &= (M^2 + 2A)^2 - 4M(MA + Q_0) \\ &= M^4 + 4(A^2 - MQ_0) = M^4 - 8. \end{aligned}$$

$\square$

6. AN EXPLICIT INFINITE FAMILY

The preceding classification shows structurally when a fixed negative value of x can support infinitely many solutions. The smallest particularly transparent example is $x = -2$.

Theorem 6.1 (An infinite family with $x = -2$). *Define positive integers A_n, B_n by*

$$(6.1) \quad A_n + B_n\sqrt{2} = (2 + \sqrt{2})(3 + 2\sqrt{2})^n, \quad n \geq 0.$$

Then for every $n \geq 0$, all four triples

$$(6.2) \quad (-2, A_n - 2, \pm B_n), \quad (-2, -A_n - 2, \pm B_n)$$

solve (1.1). These give infinitely many distinct integer solutions.

Proof. Both factors on the right side of (6.1) belong to $\mathbb{Z}[\sqrt{2}]$, so $A_n, B_n \in \mathbb{Z}$; they are positive because all coefficients in the product are positive. Taking norms gives

$$(6.3) \quad A_n^2 - 2B_n^2 = (2^2 - 2 \cdot 1^2)(3^2 - 2 \cdot 2^2)^n = 2.$$

For $x = -2$, equation (1.1) is equivalent to

$$(6.4) \quad (y + 2)^2 - 2z^2 = 2.$$

Taking $y + 2 = \pm A_n$ and $z = \pm B_n$ in (6.3) proves (6.2).

Moreover, multiplication by $3 + 2\sqrt{2}$ yields

$$(6.5) \quad A_{n+1} = 3A_n + 4B_n, \quad B_{n+1} = 2A_n + 3B_n.$$

In particular, $A_{n+1} > A_n$, so the displayed solutions are distinct as n varies. Hence there are infinitely many. $\square$

The first few values are

$$(A_n, B_n) = (2, 1), (10, 7), (58, 41), (338, 239), \dots,$$

and hence the family begins

$$\begin{aligned} &(-2, 0, \pm 1), \quad (-2, -4, \pm 1), \\ &(-2, 8, \pm 7), \quad (-2, -12, \pm 7), \\ &(-2, 56, \pm 41), \quad (-2, -60, \pm 41), \\ &(-2, 336, \pm 239), \quad (-2, -340, \pm 239), \dots \end{aligned}$$

7. CONCLUSION

Theorem 1.1 is an exact and duplication-controlled global classification: choose $m \mid y^2 + 2$, test whether $F_m(y) = my + (y^2 + 2)/m$ is a nonzero signed square, and the sign of that square determines the sign of x . Theorem 3.4 gives the precise prime-factor obstruction on $m = |x|$. For fixed m , Proposition 4.1, Theorem 4.3, and Theorem 4.6 provide an effective exhaustive procedure in every case. Finally, Theorem 6.1 proves unconditionally that the full equation has infinitely many integer solutions.